

MXCuBE Site Report

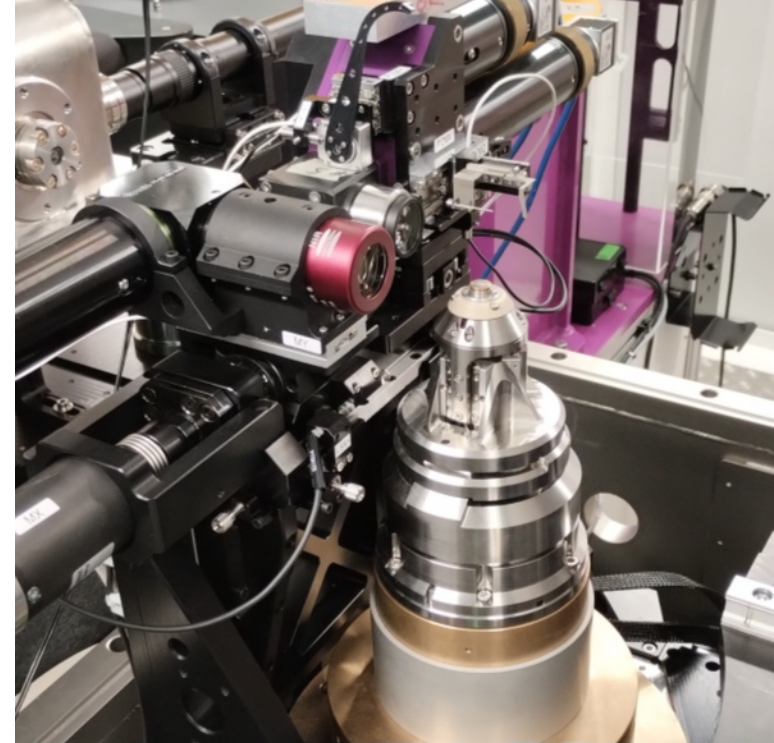
ALBA Synchrotron

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[offline] 20.05.2025 – MXCuBE/ISPyB meeting

ALBA MX Beamlines

- 2 beamlines: XALOC in production, XAIRA with inhouse users
- BL13-XALOC:
 - High-throughput tunable beamline
 - Arinax MiniDiff MD2M; Dectris Pilatus 3X 6M; IRELEC ISARA² sample-changer robot
- BL06-XAIRA:
 - Microfocus beamline
 - In-house diffractometer (Aerotech+Smaract); Dectris Eiger 9M; IRELEC ISARA² sample-changer robot
- Controls software stack:
 - Tango and Sardana, Taurus GUIs
 - LImA for cameras and detectors

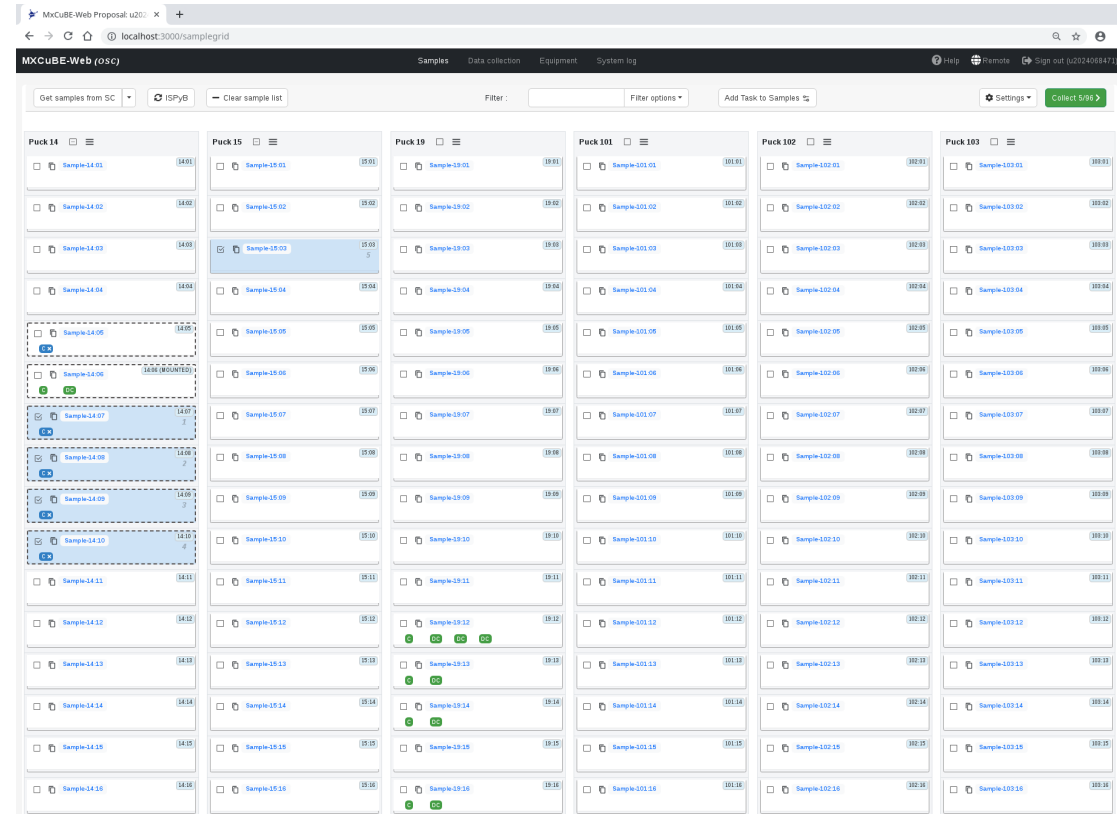


MXCuBE Status

- BL13-XALOC:
 - Python 3.9 Qt5
- BL06-XAIRA:
 - Python 3.10 Web-version, 15-month old mxcubecore
- LIMS:
 - XALOC: ISPyB with EXI
 - XAIRA: ICAT with DataPortal

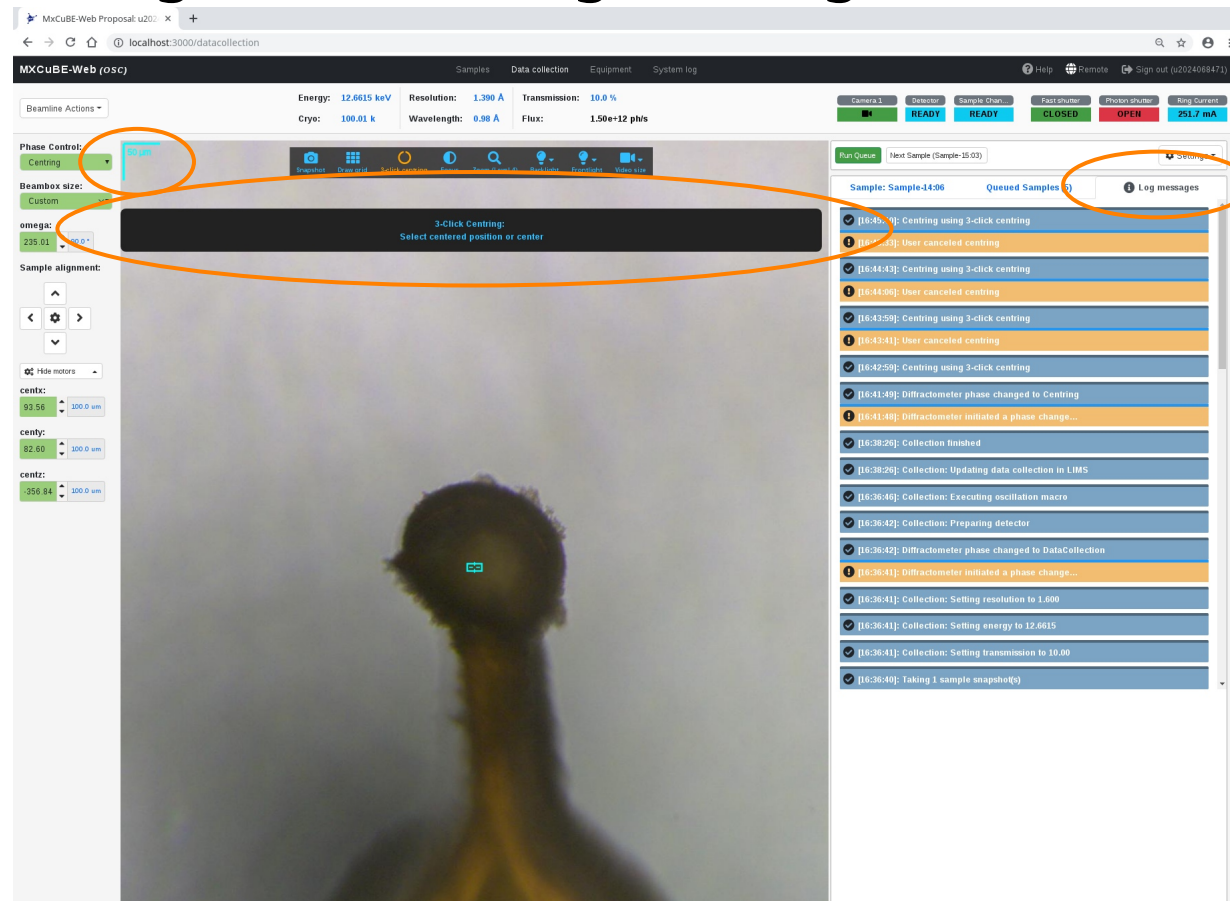
Developments Since Last Meeting

- BL13-XALOC:
 - Software “maintenance” to keep up with hardware changes and libraries version updates
 - Stability and performance improvements on transmission filters
- BL06-XAIRA:
 - Characterisation
 - Usable queue with ISARA² sample-changer
 - Better integration with ICAT
 - Helical scans
 - Improved stability i.e. bug-fixing



Developments Since Last Meeting

- BL06-XAIRA:
 - UI adaptations due to the vertical diffractometer viewport: beam scale, centring feedback, log messages were scrolled out of view



Plans For The Next Six Months

- BL13-XALOC:
 - Restart effort on the stalled update to MXCuBE-Web
- BL06-XAIRA:
 - *(finish)* X-ray centring
 - *(improve)* ICAT integration
 - Upgrade to latest community version

Acknowledgements

ID, FE and vacuum

- Pep Campmany
- Jordi Marcos
- Raquel Monge

Alignment

- Marta Llonch
- Jon Ladrera

Cryogenics/FEA

- Marcos Quispe
- Yury Nikitin
- Liudmila Nikitina

Workshop

- Jose Ferrer

MIS

- Marc Armenter
- Rodrigo Cabezas

IT

- Pau Carnicer
- Sergi Pusó
- Jordi Salabert

Technicians

- Alex Enrique

Optics and Metrology

- Josep Nicolas
- Igors Sics
- José Luis Frieiro
- Nilson Pereira

Engineering/TBM

- Nahikari González
- Alejandro Crisol
- Antonio Carballado
- Carles Colldelram

Scientists

- Judith Juanhuix
- Damià Garriga
- Isidro Crespo

Infrastructures

- Núria Martí

Electronics

- Bernat Molas
- Alberto Ruz
- Alberto Rubio
- Xevi Serra
- Sergio Astorga
- Electronics Technicians

Controls

- Jose Gabadinho
- Steven Wohl
- Oriol Vallcorba
- Fulvio Becheri
- Albert Ollé

Scientific Data Management

- Albert Castellví
- Emilio Centeno
- Joa Gómez



+ all ALBA Technical and Administrative support staff